

PROJECT 1 — Sales & Business Analytics Dashboard

Project Title

Retail Sales Analytics & Business Insights Dashboard

Why This Project?

This is one of the best projects for:

Data Analyst

Business Analyst

Analytics Intern

Reporting Analyst

BI Analyst

Recruiters instantly understand it.

Project Objective

Analyze retail/e-commerce sales data and generate:

revenue insights

customer trends

profit analysis

product performance

regional analysis

forecasting

Skills Demonstrated

Data Cleaning

Exploratory Data Analysis

SQL

Python

Pandas

Data Visualization

KPI Reporting

Dashboarding

Business Insights

Recommended Tech Stack

Core

Python

Pandas

NumPy

Matplotlib

Plotly

Streamlit

Database

MySQL or PostgreSQL

Deployment

Streamlit Cloud

Dataset Suggestions

Best Options

Superstore Sales Dataset

Walmart Sales Dataset

E-commerce Sales Dataset

Amazon Sales Dataset

Use Kaggle datasets.

Features to Include

Dashboard Pages

1. Executive Overview

Total Revenue

Profit

Orders

Growth %

Top Categories

2. Sales Trends

Monthly sales trend

Quarterly trend

Seasonal analysis

3. Customer Analytics

Top customers

Customer segmentation

Repeat customers

4. Product Insights

Best-selling products

Loss-making products

Category performance

5. Regional Analysis

State-wise sales

Region comparison

Profit heatmaps

6. Forecasting

Future sales prediction

Revenue forecasting

Advanced Features (Highly Recommended)

Add Filters

region

category

year

customer segment

Add Insights Section

Example: "Technology category generated the highest revenue but Furniture had the lowest profit margin."

Export Report

Allow dashboard users to download CSV reports.

Resume Bullet Points

Built an interactive Retail Sales Analytics Dashboard using Python, Pandas, SQL, and Streamlit.

Analyzed 10K+ sales records to identify revenue trends, customer behavior, and profit-driving products.

Developed KPI dashboards and forecasting models to support data-driven business insights.

Deployed the analytics dashboard online for real-time interactive reporting.

GitHub Structure

Repository Name

retail-sales-analytics-dashboard

Files

README.md

requirements.txt

app.py

notebook.ipynb

dataset/

screenshots/

Deployment

Best Option

Deploy using Streamlit Cloud.

This gives you:

live project link

portfolio visibility

resume credibility

Project 1: Retail Sales Analytics & Business Insights Dashboard

What this project is about

This project analyzes retail sales data and turns raw numbers into business insights. It answers questions like:

- Which products make the most profit?
- Which category is performing best?
- Which months have the highest sales?
- Which regions are underperforming?
- What should the business improve?

This is a very strong **Data Analytics** project because it shows:

- data cleaning
- exploratory data analysis
- KPI calculation
- dashboard building
- business thinking
- deployment

1) Goal of the project

The main goal is to build an interactive dashboard that helps a business understand its sales performance.

The dashboard should help a manager or analyst quickly see:

- revenue
- profit
- orders
- top products
- losing products
- region-wise performance
- monthly trends
- category-wise trends

2) Best dataset to use

The easiest and most professional choice is the **Superstore Sales Dataset** from Kaggle.

Why this dataset is good:

- it has sales, profit, quantity, discount, category, sub-category, region, ship mode, customer segment, and date fields
- it is perfect for charts and business insights
- recruiters instantly understand this type of dataset

If you want, you can later mention it as:

“Analyzed retail superstore sales data to uncover revenue trends, customer behavior, and profit drivers.”

3) What the dashboard should contain

Your dashboard should not just show charts. It should tell a story.

Page 1: Executive Summary

This is the first screen users see.

Include:

- Total Sales
- Total Profit
- Total Orders
- Average Order Value
- Profit Margin
- YoY or MoM growth if possible

Add a few small charts:

- sales trend line
- profit by category
- top 5 products
- region-wise sales map or bar chart

Page 2: Sales Trends

This page should answer:

- Are sales increasing or decreasing?
- Which month performed best?
- Are there seasonal patterns?

Include:

- line chart of monthly sales
- bar chart of sales by month
- quarter-wise comparison
- year-wise trend if dataset has multiple years

Page 3: Product Analysis

This page should answer:

- Which products sell the most?
- Which products are profitable?
- Which products are loss-making?

Include:

- top 10 products by sales

top 10 products by profit
bottom 10 products by profit
category and sub-category performance

Page 4: Regional Analysis

This page should answer:

Which region brings the highest sales?
Which region has the best profit?
Which state/city is underperforming?

Include:

sales by region
profit by region
map or heatmap if possible
state-wise comparison

Page 5: Customer Segmentation

This page should answer:

Which customer segment buys the most?
Which segment is most profitable?

Include:

sales by customer segment
profit by customer segment
average order value by segment
repeat customer insights if available

Page 6: Business Insights / Recommendations

This is the most important part for resume value.

Write 4–6 short findings like:

Technology category generates the highest revenue.
Furniture has lower profit margins due to discounts.
The West region contributes the highest sales.
Some products generate sales but have negative profit.
Certain months show strong seasonal spikes.

This is what makes the project feel like actual business analytics.

4) Features to include

To make the project stronger, include these features:

Must-have

- filters for category, region, segment, and date
- KPI cards
- charts for trends and comparisons
- clean dashboard UI
- downloadable report or CSV

Good-to-have

- interactive sidebar filters
- search by product name
- profit and sales comparison toggle
- forecast for future sales
- top insights section

Advanced but impressive

- anomaly detection for loss-making products
- forecasting with moving average or Prophet
- automatic insights generation

5) Tools and tech stack

Use this stack:

- Python** for analysis
- Pandas** for cleaning and transformation
- NumPy** for calculations
- Matplotlib / Plotly** for charts
- Streamlit** for dashboard
- Excel or CSV** as input data
- GitHub** for version control
- Streamlit Cloud** for deployment

If you want the project to feel more “analytics professional,” you can also add:

- SQL** for filtering and aggregation
- PostgreSQL/MySQL** if you want a database version

6) Step-by-step workflow

Here is how you should build it:

Step 1: Collect dataset

Download the Superstore dataset.

Step 2: Clean the data

Handle:

- missing values
- duplicate rows
- date conversion

incorrect column types

Step 3: Explore the data

Find:

- total sales
- total profit
- best categories
- worst categories
- best regions
- monthly trends

Step 4: Create KPIs

Calculate:

- total revenue
- total profit
- total orders
- average sales per order
- profit margin

Step 5: Build charts

Use visualizations for:

- time trends
- category analysis
- product analysis
- region analysis
- customer segment analysis

Step 6: Build Streamlit app

Create a dashboard with:

- sidebar filters
- summary metrics
- multiple tabs/pages
- interactive charts

Step 7: Add insights

Write human-readable conclusions, not just charts.

Step 8: Deploy

Deploy it to Streamlit Cloud and add the live link to GitHub and resume.

7) What files your GitHub repo should have

Your GitHub repository should look clean and professional.

Suggested structure:

```
retail-sales-analytics-dashboard/  
├── app.py  
├── data/  
│   └── superstore.csv  
├── notebooks/  
│   └── analysis.ipynb  
├── assets/  
│   └── screenshots/  
├── requirements.txt  
├── README.md  
└── utils.py
```

8) What to write in README

Your README should include:

- project title
- problem statement
- dataset source
- tools used
- project workflow
- dashboard screenshots
- key insights
- deployment link
- future improvements

This matters a lot because recruiters often check GitHub fast.

9) Resume value of this project

This project helps you show:

- business understanding
- analytical thinking
- dashboard creation
- data storytelling
- deployment skills
- GitHub professionalism

That is exactly what a Data Analytics role needs.

10) Best project title options

You can name it in a more polished way, such as:

Retail Sales Analytics Dashboard
Sales Performance Insights Dashboard
Business Intelligence Dashboard for Retail Sales
Interactive Sales Analytics System

The best one for resume is probably:

Retail Sales Analytics & Business Insights Dashboard

11) Resume bullet examples

You can later use bullets like these:

Built an interactive retail sales analytics dashboard using Python, Pandas, Plotly, and Streamlit.
Analyzed sales, profit, and customer segment data to uncover business trends and underperforming areas.
Designed KPI dashboards and interactive filters for regional, category, and product-level insights.
Deployed the dashboard online and documented the project on GitHub with complete analysis and screenshots.

12) What makes this project strong

This project is strong because it is:

practical
easy to explain in interviews
visually impressive
relevant to Data Analyst jobs
deployable
GitHub-friendly

It also gives you good talking points for:

project explanation
dashboard walkthrough
data cleaning decisions
business insights
deployment

13) What you should do next

The best next step is to decide the exact dataset and dashboard style.

My recommendation:

use the **Superstore dataset**
build it in **Streamlit**
deploy it on **Streamlit Cloud**

make it look clean and business-like

Summary: This project should be an interactive retail sales dashboard with KPIs, trends, product analysis, region analysis, and business insights. It is one of the strongest beginner-to-intermediate Data Analytics projects for your resume.

DATASETS:

<https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting>

<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

<https://www.kaggle.com/datasets/thedevastator/unlock-profits-with-e-commerce-sales-data>

[https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset?
select=Global_Superstore2.csv](https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset?select=Global_Superstore2.csv) [Using]

<https://www.kaggle.com/datasets/yashyennewar/product-sales-dataset-2023-2024>

Smart Next Step

Your roadmap is actually very good:

Current Phase

- ✓ Data cleaning
- ✓ KPI analysis
- ✓ Business analytics
- ✓ Visualization system
- ✓ Dashboard assets
- ✓ Summary tables

Final Phase

- ▢ Forecasting
- ▢ Strategic recommendations
- ▢ Decision intelligence
- ▢ AI insights
- ▢ Predictive analytics

That progression is exactly how real BI systems evolve.

So the correct path is:

✓ NOW

- Finalize dashboard structure
- Clean visuals
- Remove weak/redundant charts
- Organize sections professionally
- Create polished dashboards
- Push to GitHub
- Deploy/host the project
- Prepare portfolio-ready presentation



LATER

- Forecasting
- AI insights
- Predictive analytics
- Decision intelligence
- Recommendation systems

Phase 1 — Final Polish (NOW)

Focus on:

- dashboard layout
- visual consistency
- KPI cards
- responsiveness
- navigation
- dashboard storytelling
- performance optimization
- export-ready assets

Phase 2 — Deployment (NOW)

Do:

- organize repo structure
- clean README
- add screenshots/GIFs
- document features
- upload to GitHub
- deploy dashboard

Possible hosting:

- Streamlit Cloud
- Render
- Hugging Face Spaces
- Vercel (frontend)
- GitHub Pages (static)
- Power BI Service

Phase 3 — Portfolio Packaging (NOW)

Add:

- architecture diagram
- feature list
- business objectives
- insights generated
- dashboard screenshots
- tech stack
- folder organization

This massively increases project quality.

Phase 4 — Intelligent Layer (LATER)

Only after deployment:

- forecasting
- predictive analytics
- anomaly detection
- recommendation engine
- AI-generated insights
- decision intelligence

At that point, it becomes:

Retail Analytics + AI Intelligence Platform

which can even become:

- research-level
- startup-level
- enterprise-level